INCH-POUND

MIL-S-20708/70D
13 March 1989

SUPERSEDING
MIL-S-20708/70C
3 March 1969

MILITARY SPECIFICATION SHEET
SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 37TRX4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708.

TABLE I. REQUIREMENTS

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	115	volts	±1%
Primary Current	1885	milliamps	maximum
Primary Power	20	watts	maximum
Impedance:			
Zss	3.0-4.0	ohms	min.-max.
Impedance Angle:			
Zss	66.0-70.0	degrees	min.-max.
Transformation Ratio	0.783	-----	±2%
Phase Shift (Lead)	1.0	degrees	maximum
Electrical Error	8.0	minutes	min.-max.
Receiver Error	36	minutes	maximum
Null Voltage:			
Total	170	millivolts	maximum
Fundamental	35	millivolts	maximum
Synchronizing Time:			
30° ±2°	1.0	seconds	maximum
Synchronizing Time:			
177° ±2°	2.0	seconds	maximum
Torque Gradient	0.9	ounce-inches per degree	minimum
Radial Play	0.0008	inches	maximum
End Play	0.0030	inches	±0.0020
Temperature Rise	60	degrees (C)	maximum
Variation of Voltage (+10%) and Freq. (-5%)	32.5	watts	maximum

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TABLE II. Military part number variant characteristics.

Military Part No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U +.0000 −.0002
M20708/70-01A	.700	.730	4.030	Splined	Terminal Board	.2405

1/Military part number M20708/70-01A shall be the NATO standard.

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Military part number cross-references.

SUPERSEDED MILITARY PART NUMBERS	NEW MILITARY PART NUMBERS
MIL-S-20708/70C	
M20708/70-001	M20708/70-01A

NOTE: All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification. Part number changes do not affect form, fit, or function of the synchros listed therein.

Custodians:
Navy-AS

Review Activities:
NAVY-EC
DLA-ES

User Activities:
Navy-MC,CG

Preparing Activity:
Navy-AS

Project 5990-0349-76

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